

Dynamic Programming Notes: Paint Fence

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SUPPOSE

$n = 5$

$k = 3$

Fence0

Fence1

Fence2

Fence3

Fence4

Each fence has

3 colors

Question:

When painting the current fence, We only care about
Last two fences.

Because only they can create
XXX

In Paint House
Adjacent fences cannot have same color.

Here
No more than TWO consecutive fences
can have the same color.

So
RR ✓
RRR ✗

Three consecutive equal colors are forbidden.

Cases

Case 1

Last two fences are
Different

Example
R G

Let's call this

diff

Case 2

Last two fences are
Same

Example
R R

Let's call this

same

DP State

Let `same[i]` =

Number of ways such that
Last two fences are same.

Let `diff[i]` =

Number of ways such that
Last two fences are different.

Base Case

One fence

Choose any color

`same` = k

`diff` = 0

There is no pair yet.

Two fences

Same

RR

GG

BB

Total

`same` = k

Different

Choose first color

k

Choose second

$k-1$

Therefore

`diff` = $k*(k-1)$

Transitions

To make **Same** last two fences,
previous state must be **Different**

Example

R G

Paint G becomes R G G Valid.

But if previous already had R R adding R gives R R R Invalid.

Compute diff

Current fence must use
Any color except previous.

How many choices?

$k-1$

Previous configuration can be
same or diff

Transition

```
newSame = diff;
```

```
newDiff = (same+diff)*(k-1);
```

Implementation

```
class Solution {
public:
    int numWays(int n, int k) {

        if(n == 1)
            return k;

        // same
        // Number of ways such that
```

```
// last two fences have same color.
long long same = k;

// diff
// Number of ways such that
// last two fences have different colors.
long long diff = 1LL * k * (k - 1);

// Paint remaining fences.
for(int i = 3; i <= n; i++) {

    long long newSame = diff;
    long long newDiff = (same + diff) * (k - 1);

    same = newSame;
    diff = newDiff;
}

// Last two fences can end
// either same or different.
return same + diff;
}
};
```